



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.9

Represent Polygons on the Coordinate Plane

Lesson 7-7 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

Language Objective: I can explain how I found a distance using the words vertices, coordinates, and absolute value.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

vertex

polygon

perimeter

distance between vertices

scale

Side length



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A corner point of a polygon, named by an ordered pair, is a

2

A closed figure made of line segments is a

3

The total distance around a polygon is its

4

When two vertices share a coordinate, subtract the other and take the

5

What one grid unit stands for in the real situation is the

6

The distance between two joined vertices is a .

Write About the Math

Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?

¿A dónde van los otros dos vértices — y cómo trabajan juntas las coordenadas conocidas, la escala y el área para decírtelo?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **vertex**
vértice

☐ **polygon**
polígono

☐ **perimeter**
perímetro

☐ **distance between vertices**
distancia entre vértices

☐ **scale**
escala

Model: When two vertices share the same *y*-coordinate, the segment between them is horizontal, so its length comes from the *DIFFERENCE* in the coordinate that changes — the *x*-coordinates. — A strong answer explains that the *SHARED* coordinate tells you the direction of the side, and the *DIFFERING* coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- **My evidence is** ____, **so** ____.

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Notes 6:** Side length
7. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from $y = 3$ down to $y = -3.75$ the distance is $3 + 3.75 = 6.75$ units, not $3.75 - 3 = 0.75$. Distances that cross zero are the whole reason the absolute value is there.*
8. **Try It 1:**
9. **Try It 2:**